

BUBBLE SORT:

Bubble Sort is one of the simplest sorting algorithms. It repeatedly compares two adjacent elements and swaps them if they are in the wrong order. This process continues until the array is completely sorted. In each pass, the largest unsorted element "bubbles up" to its correct position.

Algorithm:

1. Read the number of elements.
2. Enter the array elements.
3. Compare each pair of adjacent elements.
4. If the left element is greater than the right element, swap them.
5. Continue comparing until the end of the array.
6. Repeat the process for all remaining unsorted elements.
7. Display the sorted array.

Time Complexity:

Case	Time Complexity
Best Case	$O(n)$
Average Case	$O(n^2)$
Worst Case	$O(n^2)$

Space Complexity:

Type	Space Complexity
Bubble Sort	$O(1)$

Output:

```
Enter the number of elements:  
5  
Enter the elements:  
2  
4  
1  
5  
3  
Sorted array:  
1 2 3 4 5  
Process finished with exit code 0
```